Project Task Description

Project Title: Host a Secure Static Website Using Amazon S3, CloudFront and Route 53

Objective

Deploy a company marketing website to AWS so that it is served to the public over HTTPS from a global content delivery network, while the S3 bucket holding the files remains completely private. Create appropriate IAM identities and work from those rather than from the root account.

Tasks

1. Secure the Account and Create Working Identities

- Enable MFA on the root user and stop using it for day-to-day work.
- Create a Cost budget of $10 per month with alerts at 80% and 100% of actual spend, and at 100% of forecasted spend, before deploying any resources.
- Navigate to IAM and create a group named Developers.
- Attach the AmazonS3FullAccess policy to the group. Attach nothing else.
- Create an IAM user for yourself with console access and add it to the Developers group.
- Enable MFA on the IAM user.
- Copy the IAM sign-in URL from the IAM dashboard and save it. Sign out, then sign back in as the IAM user.
- Open the EC2 console and record the access denied message. Open the S3 console and confirm it works.

You are authenticated in both cases. You are authorized in only one. This distinction is assessed.

2. Create an S3 Bucket and Upload the Website

- Navigate to S3 and click Create bucket.
- Enter a globally unique, DNS-compliant bucket name: lowercase only, no spaces or underscores, hyphens permitted.
- Choose the AWS Region agreed by your group and do not change it again.
- Leave Block all public access enabled.
- Upload index.html, about.html, error.html, style.css and logo.png at the root level of the bucket.
- Open the Object URL for index.html in a browser and record the AccessDenied response.

3. Enable Static Website Hosting

- Open the Properties tab and scroll to the bottom to Static website hosting.
- Click Edit and select Enable.
- Set Index document to index.html and Error document to error.html.
- Save changes and copy the Bucket website endpoint.

The website endpoint uses http, not https. Record why this matters for the next task.

4. Create a CloudFront Distribution with Origin Access Control

- Navigate to CloudFront and click Create distribution.
- Enter a distribution name and select Single website or app as the distribution type.
- Leave the Route 53 managed domain field empty unless you own a registered domain.
- For Origin type select Amazon S3, then use Browse S3 to select your bucket.
- Under Settings select Use recommended origin settings, which configures Origin Access Control and updates the bucket policy automatically.
- Choose Do not enable security protections.
- Create the distribution and wait for the status to leave Deploying.
- Open the distribution settings, click Edit, and set Default root object to index.html.

The wizard does not ask for a default root object. Without it, the bare distribution URL returns an error while the /index.html path succeeds.

5. Lock the Origin Bucket Down

- Open the bucket Permissions tab and read the bucket policy that CloudFront generated.
- Compare its Principal and Condition with a public-read policy and record the difference.
- Delete any earlier public-read policy statement, keeping only the CloudFront statement.
- Re-enable Block all public access and confirm.
- Verify that the CloudFront URL loads over HTTPS, and that both the S3 website endpoint and the S3 object URL return an access denied response.

6. Point a Custom Domain Using Route 53 (Optional)

- Create a public hosted zone for a domain you own and delegate the four name servers at your registrar.
- Request a public certificate in AWS Certificate Manager in the us-east-1 Region.
- Validate the certificate using DNS validation.
- Add the domain under Alternate domain names on the CloudFront distribution and attach the certificate.
- Create an A record with Alias enabled, pointing to the CloudFront distribution.

A hosted zone is charged at $0.50 per month for as long as it exists, including when empty.

7. Tear Down

- Disable the CloudFront distribution, wait for the change to complete, then delete it.
- Empty the S3 bucket, then delete it.
- Delete the Route 53 hosted zone if you created one.
- Retain your IAM user, the Developers group and your budget alert for later projects.

Submission Requirements

Learners must submit:

- The CloudFront distribution URL
- The screenshots listed in the Deliverables table below

Resources:

AWS Documentation: Refer to AWS documentation for detailed guidance on each service used in this project.

Walkthrough Video: Link (add your group's recording here)

Deliverables

Deliverable	Evidence Required
1. IAM Identity Configuration	Screenshot of the Developers group showing the attached policy and your IAM user listed as a member. *File name: P1_01_IAMGroup_FullName*
2. Least Privilege Demonstrated	Screenshot of the access denied message shown when opening the EC2 console while signed in as your IAM user. *File name: P1_02_AccessDenied_FullName*
3. Bucket Creation and DNS-Compliant Naming	Screenshot showing the bucket name following DNS rules: lowercase only, no spaces or underscores, uses hyphens. *File name: P1_03_BucketName_FullName*
4. Website File Structure	Screenshot of the Objects tab showing index.html at the root level of the bucket alongside the remaining site files. *File name: P1_04_FileStructure_FullName*
5. Static Website Hosting Enabled	Screenshot of the Static website hosting panel showing Enabled, the index and error documents, and the bucket website endpoint. *File name: P1_05_WebsiteHosting_FullName*
6. CloudFront Distribution Deployed	Screenshot of the distribution showing status Enabled and Default root object set to index.html. *File name: P1_06_CloudFront_FullName*
7. Public Website, Private Bucket	Two screenshots: the CloudFront URL loading over HTTPS with the padlock visible, and the direct S3 URL returning an access denied response. *File name: P1_07_PrivateOrigin_FullName*
8. Resource Cleanup	Screenshot showing the CloudFront distribution deleted or disabled and the S3 bucket removed. *File name: P1_08_Cleanup_FullName*

Refer to the table and submit your screenshots as evidence for each deliverable to this form via the link below.
Link to the form: Click here to access the submission form.